

Enterprise WhatsApp Bulk Messaging & Campaign Management Platform

Project Overview

Project Name

Enterprise WhatsApp Bulk Messaging & Campaign Management Platform

Project Type

SaaS-Based Web Application

Development Team

- Frontend Team (Next.js)
 - Backend Team (Rust + Axum)
 - Automation Team (n8n Workflows)
 - DevOps Team
 - QA Team
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Problem Statement

Businesses, educational institutions, marketing agencies, political organizations, and service providers require a reliable and scalable solution to communicate with thousands of users through WhatsApp.

Current communication methods suffer from several limitations:

- Manual message sending is time-consuming.
- No centralized campaign management.
- Difficult customer segmentation.
- Lack of automation workflows.
- No detailed analytics.
- Poor scalability.
- Difficult management of multiple WhatsApp numbers.
- Limited scheduling and follow-up capabilities.

The objective is to build a centralized platform that enables organizations to manage WhatsApp communications efficiently, automate workflows, monitor campaign performance, and scale messaging operations.

Business Objectives

The platform should allow users to:

- Connect WhatsApp accounts.
 - Manage customer databases.
 - Create and manage campaigns.
 - Send bulk WhatsApp messages.
 - Schedule campaigns.
 - Automate customer communication.
 - Track campaign performance.
 - Manage team members.
 - Generate reports and analytics.
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Technology Stack

Frontend

Framework

- Next.js 15
- TypeScript

UI Framework

- Tailwind CSS
- ShadCN UI

Animations

- Framer Motion

State Management

- Zustand

API Layer

- TanStack Query
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Backend

Language

- Rust

Framework

- Axum

Authentication

- JWT Authentication
- Refresh Tokens

Security

- RBAC (Role-Based Access Control)
 - API Rate Limiting
 - Data Encryption
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Database

Primary Database

- PostgreSQL

Caching

- Redis

Object Storage

- S3 Compatible Storage
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Workflow Automation

Platform

- n8n

Responsibilities

- Bulk messaging automation
 - Campaign execution
 - Scheduled workflows
 - Auto-response workflows
 - Lead nurturing automation
 - Event-based messaging
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User Roles

Super Admin

Permissions

- Manage entire platform
 - Manage organizations
 - Manage subscriptions
 - View system analytics
 - Monitor workflow execution
 - Manage user access
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Business Admin

Permissions

- Manage organization
 - Manage team members
 - Manage contacts
 - Create campaigns
 - Manage workflows
 - View reports
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Team Member

Permissions

- Create campaigns
 - Manage assigned contacts
 - Access assigned conversations
 - View reports
-

Core Modules

Module 1: Authentication & Authorization

Features

- User Registration
- User Login
- Forgot Password

- Email Verification
 - Two-Factor Authentication
 - JWT Authentication
 - Role-Based Access Control
 - Team Invitations
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Module 2: Organization Management

Features

- Create Organization
 - Manage Teams
 - Assign Roles
 - Manage Permissions
 - Organization Settings
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Module 3: WhatsApp Account Management

Features

- QR Code Authentication
 - Connect WhatsApp Number
 - Multi-Number Support
 - Connection Monitoring
 - Reconnection Handling
 - Account Status Tracking
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Module 4: Contact Management

Features

- Import Contacts via CSV
- Import Contacts via Excel
- Manual Contact Creation
- Contact Tags
- Contact Groups
- Dynamic Segmentation

Example Segments

- Leads
- Customers
- Students
- Vendors

- Supporters
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Module 5: Campaign Management

Features

- Create Campaign
- Edit Campaign
- Save Draft
- Schedule Campaign
- Immediate Campaign Execution
- Campaign Cloning

Campaign Types

- Promotional
 - Transactional
 - Notifications
 - Reminders
 - Surveys
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Module 6: Bulk Messaging Engine

Features

- High-volume message delivery
- Batch processing
- Queue management
- Message personalization
- Delivery tracking

Example Template

```
Hello {{name}},  
  
Your order {{order_id}} has been successfully processed.  
  
Thank you.
```

Module 7: Automation Workflows

Powered By

n8n

Features

- Event-Based Triggers
- Follow-Up Sequences
- Lead Automation
- Reminder Automation
- Conditional Logic
- Workflow Monitoring

Example Workflow

```
graph TD; A[Lead Created] --> B[Welcome Message]; B --> C[Wait 1 Day]; C --> D[Follow-Up Message]; D --> E[Wait 3 Days]; E --> F[Notify Sales Team];
```

Module 8: Template Management

Features

- Create Templates
- Store Templates
- Categorize Templates
- Dynamic Variables
- Template Library

Module 9: Inbox & Conversations

Features

- Receive Customer Messages

- Chat Interface
 - Search Conversations
 - Assign Conversations
 - Conversation History
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Module 10: Analytics Dashboard

Metrics

- Messages Sent
- Messages Delivered
- Messages Failed
- Read Rate
- Response Rate
- Campaign Performance

Dashboard Widgets

- Daily Usage
 - Weekly Usage
 - Monthly Usage
 - Campaign Trends
 - User Activity
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Module 11: Scheduling System

Features

- One-Time Scheduling
- Recurring Scheduling
- Timezone Support
- Scheduled Campaign Monitoring

Schedule Types

- Daily
 - Weekly
 - Monthly
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Module 12: Subscription & Billing

Features

- Subscription Plans

- Usage Monitoring
- Billing History
- Invoice Generation
- Plan Upgrades

Example Plans

- Starter
 - Professional
 - Enterprise
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Non-Functional Requirements

Performance

Target

- Support 100,000+ Contacts
 - Support 10,000+ Messages Per Campaign
 - API Response Time Below 300ms
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Scalability

Requirements

- Horizontal Scaling
 - Containerized Deployment
 - Queue-Based Architecture
 - Microservice Ready Design
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Security

Requirements

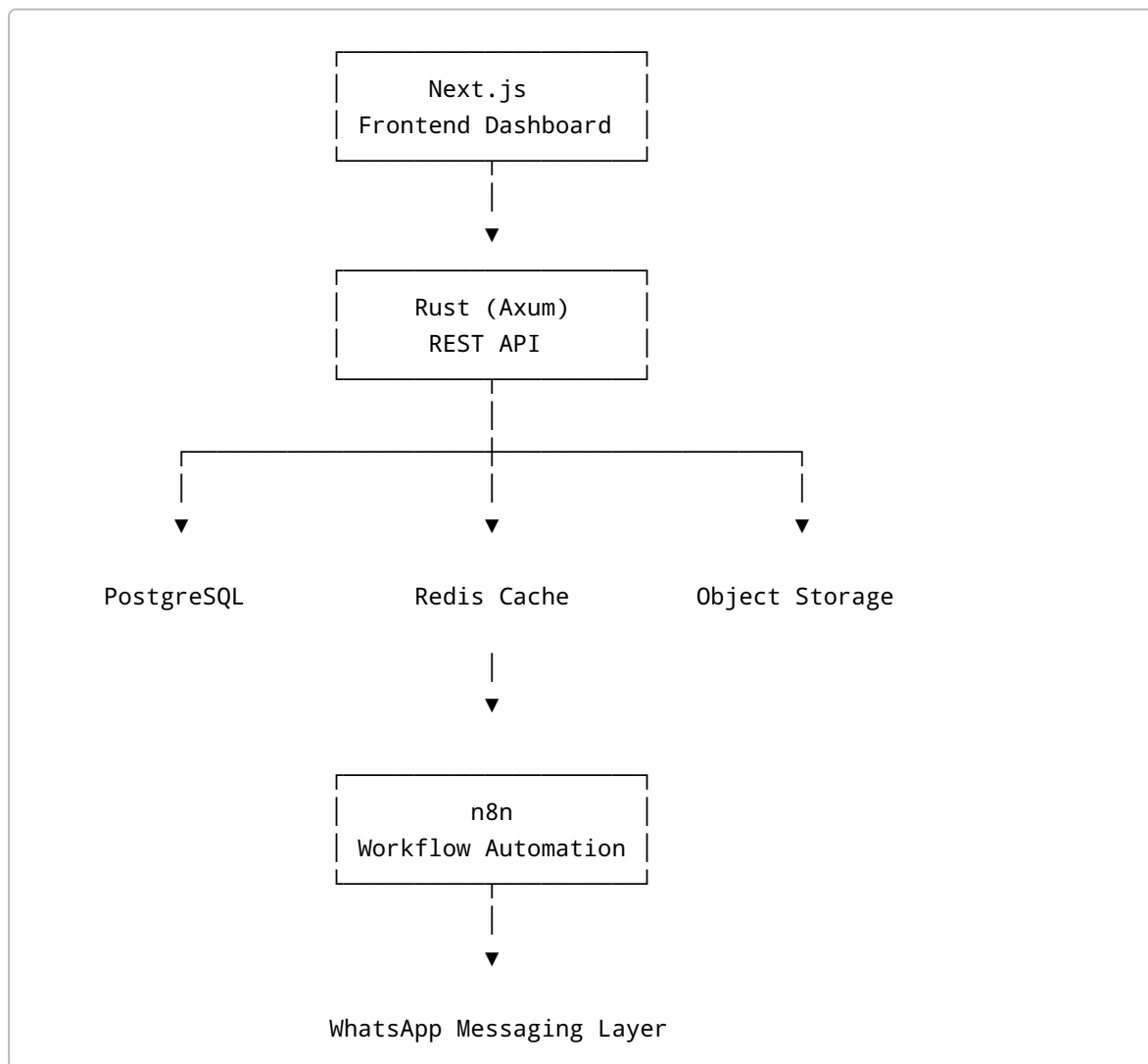
- JWT Authentication
 - Encrypted Data Storage
 - Secure API Access
 - Audit Logging
 - Role-Based Access Control
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Reliability

Requirements

- 99.9% Uptime
 - Workflow Retry Mechanism
 - Automatic Recovery
 - Database Backups
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High-Level System Architecture



Development Roadmap

Phase 1 - MVP

Deliverables

- Authentication System
 - WhatsApp Connection
 - Contact Management
 - Campaign Creation
 - Bulk Messaging
 - Basic Analytics
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Phase 2

Deliverables

- n8n Integration
 - Workflow Automation
 - Team Management
 - Scheduling System
 - Template Management
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Phase 3

Deliverables

- SaaS Billing System
 - Advanced Analytics
 - AI Features
 - Multi-Tenant Architecture
 - Enterprise Features
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Future AI Features

AI Message Generation

Generate campaign content automatically.

AI Contact Segmentation

Automatically group users based on behavior.

AI Send-Time Optimization

Predict the best time to send messages.

AI Reply Suggestions

Generate intelligent customer response suggestions.

AI Campaign Analysis

Provide recommendations for improving campaign performance.

Success Criteria

The project will be considered successful if:

- Users can connect WhatsApp accounts seamlessly.
- Campaigns can be sent to large contact lists reliably.
- Automation workflows run without failure.
- Organizations can manage teams and contacts efficiently.
- Analytics provide actionable insights.
- Platform remains scalable and secure under high load.